Dataset — Sleep-EDF Expanded

Overview

Property	Value
Source	PhysioNet Sleep-EDF Expanded
Type	Polysomnography (PSG) recordings
Subjects	4 (development subset)
Recordings	4 nights
Total epochs	~11,128 (full cohort)
Epoch length	30 seconds
Sampling rate	100 Hz
Target classes	5 (AASM standard)

Data Source

PhysioNet Sleep-EDF Expanded Database

- URL: https://physionet.org/content/sleep-edfx/1.0.0/ (https://physionet.org/content/sleep-edfx/1.0.0/)
- License: Open Access
- Citation: Goldberger et al., 2000

The Sleep-EDF dataset contains polysomnography recordings from healthy subjects. Each recording includes:

- EEG signals (multiple channels)
- EOG (electrooculography)
- EMG (electromyography)
- Hypnogram annotations (expert-scored sleep stages)

Channels Used

Channel	MNE Name	Purpose
EEG 1	EEG Fpz-Cz	Brain electrical activity (frontal)
EEG 2	EEG Pz-Oz	Brain electrical activity (parietal-occipital)
EOG	EOG horizontal	Eye movement detection
EMG	EMG submental	Muscle activity

Why These Channels?

EEG Fpz-Cz:

- Primary channel for sleep stage classification
- Captures delta waves (N3), sleep spindles (N2), alpha rhythm (Wake)

EEG Pz-Oz:

- Complementary posterior EEG
- Helps distinguish N2 from N3 stages

EOG:

- Critical for REM detection (rapid eye movements)
- Helps identify Wake state (voluntary eye movements)

EMG:

- Muscle tone changes across sleep stages
- Low tone in REM, higher in Wake
- Helps distinguish Wake from N1

Sleep Stages (AASM Standard)

Code	Stage	Description	Typical EEG Features
0	Wake (W)	Awake state	Alpha rhythm (8-13 Hz), beta activity
1	N1	Light sleep transition	Theta waves (4-8 Hz), vertex sharp waves
2	N2	Stable light/intermediate sleep	Sleep spindles, K-complexes
3	N3	Deep/slow-wave sleep	Delta waves (0.5-4 Hz), high amplitude
4	REM	Rapid eye movement sleep	Mixed frequency, low amplitude, rapid eye movements

Class Distribution

The dataset exhibits extreme class imbalance (untrimmed full-night recordings):

```
Wake:   68.8% of epochs (28,219) — majority class
N1:      3.4% of epochs (1,388)  — minority class
N2:     16.4% of epochs (6,718)
N3:      5.0% of epochs (2,070)
REM:     6.4% of epochs (2,642)
```

> ***Why Wake is 68.8%:*** *These are untrimmed full-night recordings that include pre-sleep and post-sleep wake periods. Standard practice in the literature trims to ±30 min around sleep onset/offset, which would rebalance to ~10-15% Wake. This pipeline uses untrimmed recordings for now.*

Impact on training: The pipeline uses:

- N1/REM class weighting (2x) to address minority classes
- Macro F1 and Cohen's κ as primary metrics (accuracy-insensitive to imbalance)

Data Preprocessing Pipeline

Raw Signal → Clean Epochs

```
Raw PSG (100 Hz, 4 channels)
    │
    ▼
┌─────────────────────────────────┐
│ 1. Bandpass Filter              │
│    - 4th order Butterworth      │
│    - 0.5 Hz ≤ f ≤ 35 Hz         │
│    - Removes: baseline drift,   │
│      high-frequency noise       │
└─────────────────────────────────┘
              │
              ▼
┌─────────────────────────────────┐
│ 2. Notch Filter                 │
│    - IIR notch at 50 Hz         │
│    - Q factor: 30               │
│    - Removes: power-line noise  │
└─────────────────────────────────┘
              │
              ▼
┌─────────────────────────────────┐
│ 3. Z-Score Normalization        │
│    - Per-channel, per-epoch     │
│    - x_norm = (x - μ) / σ       │
│    - Ensures zero mean, unit    │
│      variance                   │
└─────────────────────────────────┘
              │
              ▼
┌─────────────────────────────────┐
│ 4. Artifact Quality Control     │
│    - Clipping detection         │
│      (|x| > 8.0 σ)              │
│    - Flatline detection         │
│      (std < 0.05)               │
│    - NaN/Inf detection          │
│    - Flag rate: ~2%             │
└─────────────────────────────────┘
              │
              ▼
    Clean Epochs (30s × 4ch × 100Hz)
    Shape: [n_epochs, 4, 3000]
```

Dataset Splits

Subject-Level Splitting

To prevent data leakage, splits are performed at the **subject level**, not at the epoch level.

Split	Subjects	Recordings	Purpose
Train	70%	~55 subjects	Model training
Validation	15%	~12 subjects	Hyperparameter tuning, checkpoint selection
Test	15%	~12 subjects	Final evaluation

Key principle: No subject appears in multiple splits. This ensures the model generalizes to unseen subjects.

Split Assignment

```
from sklearn.model_selection import train_test_split

subjects = sorted(manifest["subject_id"].unique())
train_subj, temp_subj = train_test_split(subjects, test_size=0.30, random_state=42)
val_subj, test_subj = train_test_split(temp_subj, test_size=0.50, random_state=42)
```

Data Contract

The preprocessing pipeline outputs the following artifacts:

Manifest File

```
subject_id,night,psg,hypnogram,split
SC4001,0,data/raw/sleep_edf/SC4001E0-PSG.edf,data/raw/sleep_edf/SC4001EC-Hypnogram.edf,train
SC4002,0,data/raw/sleep_edf/SC4002E0-PSG.edf,data/raw/sleep_edf/SC4002EC-Hypnogram.edf,train
```

Cache Files

```
# Per subject-night .npz file
{
    "epochs": np.ndarray,      # [n_epochs, 4, 3000] float32
    "labels": np.ndarray,      # [n_epochs] int64, values 0-4
    "qc_flag": np.ndarray,     # [n_epochs] bool
    "fs": float,               # 100.0
    "subject_id": str,         # "SC4001"
    "night": int,              # 0
    "split": str               # "train" | "val" | "test"
}
```

Cache Index

```
subject_id,night,split,n_epochs,n_flagged,cache_path
SC4001,0,train,2649,50,data/cache/SC4001_night0.npz
SC4002,0,train,2829,56,data/cache/SC4002_night0.npz
```

Data Governance

Integrity Checks

The pipeline performs these validation steps:

1. **PSG-Hypnogram pairing:** Each PSG file must have a matching hypnogram
2. **File existence:** All referenced files must exist on disk
3. **No duplicate subject-nights:** Each (subject_id, night) pair is unique
4. **Channel availability:** All 4 required channels must be present
5. **Label validation:** All labels must be in {0, 1, 2, 3, 4}
6. **No NaN/Inf:** Processed signals must be finite

Reproducibility

Parameter	Value	Purpose
Random seed	42	Reproducible subject splits
Filter order	4	Deterministic filtering
Normalization	Per-epoch z-score	Consistent across runs

Dataset Limitations

1. **Development subset expanded:** The pipeline now uses 15 subjects from Sleep-EDF Expanded. Full deployment requires 183 subjects.
2. **Healthy subjects only:** No pathological sleep patterns (apnea, narcolepsy, etc.)
3. **Single night per subject:** Limited intra-subject variability
4. **Class imbalance:** N1 and REM are underrepresented
5. **Annotation granularity:** 30-second epochs may miss brief events

---

Citation

```
@article{goldberger2000physiobank,
  title={PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals},
  author={Goldberger, Ary L and others},
  journal={Circulation},
  volume={101},
  number={23},
  pages={e215--e220},
  year={2000}
}
```

Last updated: August 2026 Project: Neuromorphic Sleep Stage Scoring — VIT Bhopal University